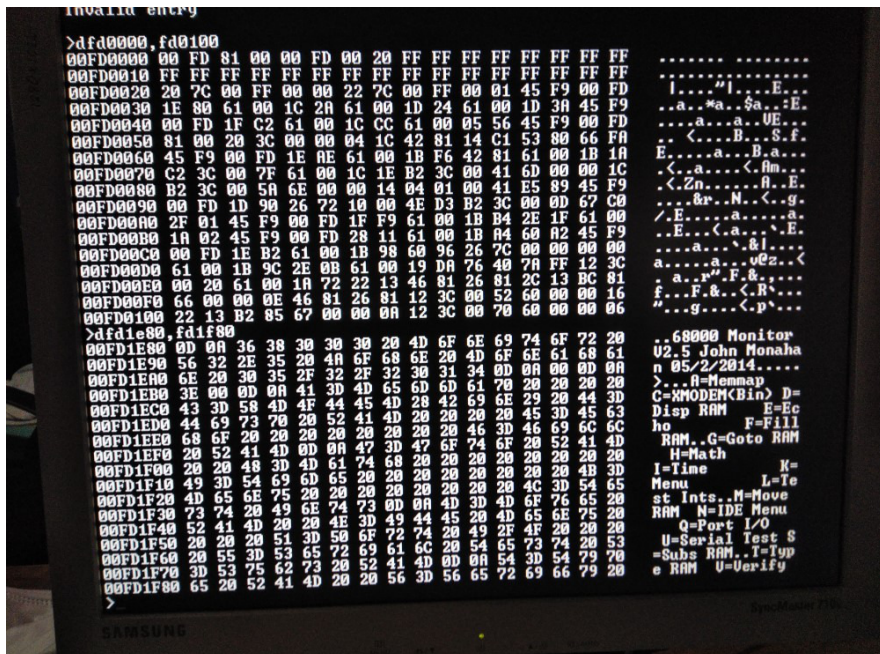


COMMISSIONING of the CPU 68000 V1.03B S100Computers



Introduction

The objective of this document is to build a tool oriented to do a easy commissioning of the CPU 68000-68010 V1.03 S-100 board.

Based on an early experience and being a way to analyze the S-100 hardware functionalities, the content can not be used to find and accurate information of the board. This is only a navigation chart which puts in order the compilation done of a couple of years from the same information that can be found in S100 Computers web and forum community.

Structure of the document:

- Initial questions
- Table of Configuration Point
- Schematic pages, linked from the table
- A final Conclusion
- General Lay-out of the board.

Initial questions

A couple of first question shall be solved before to start the commissioning:

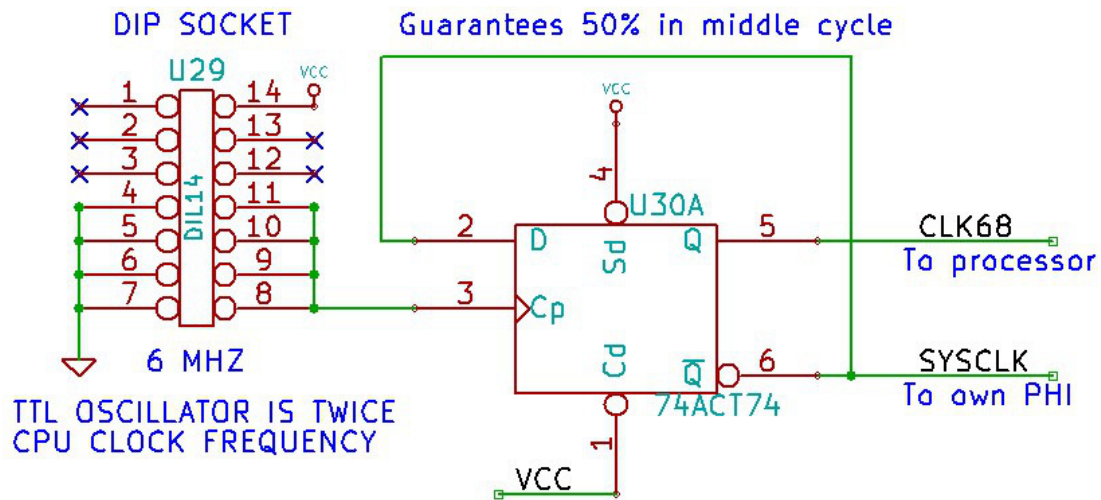
In the knowledge that U30A is a “divider by two” circuit, the oscillator can be selected as a **double** of the frequency that we want to applied in the board.

Before starting is also needed to define the type of functionality to be configured, as **Temporary**, **Permanent** or **Alone** Master.

Finally, this couple of pages shall be used to introduce to the **how to find** and **how to decide** to select a correct strap position.

Choosing a Frequency

GENERAL CLOCK CIRCUIT

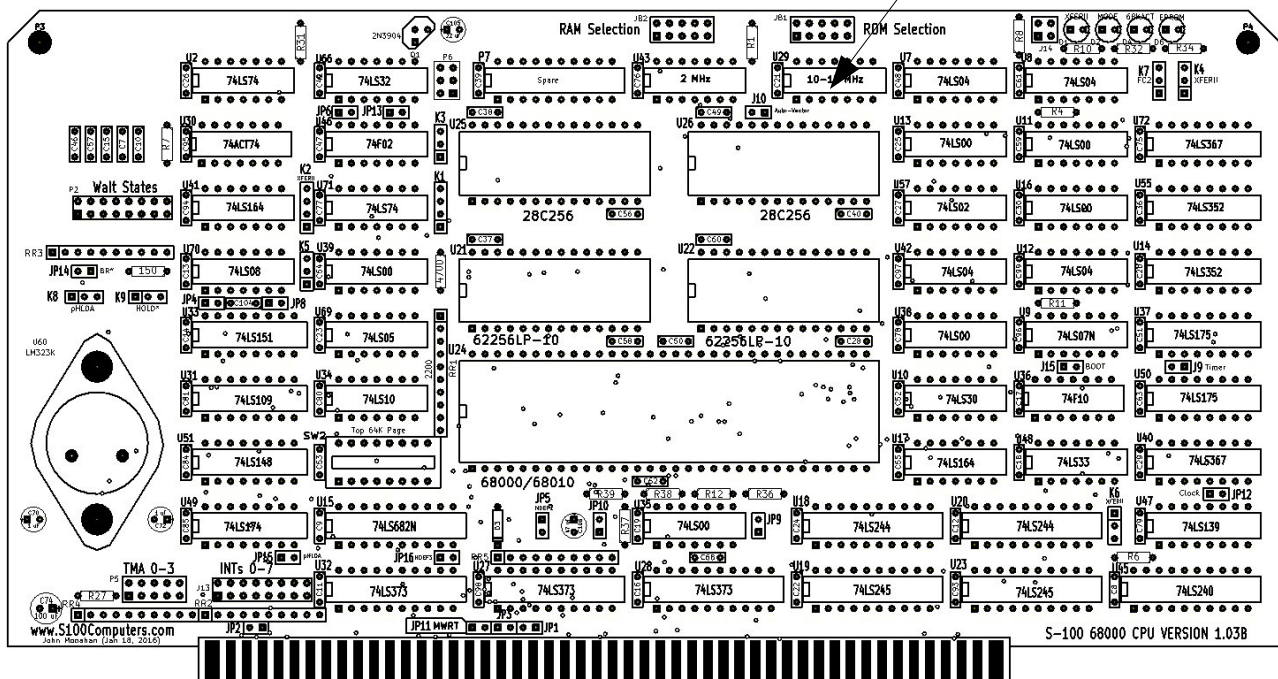


Two restrictions at the same time:

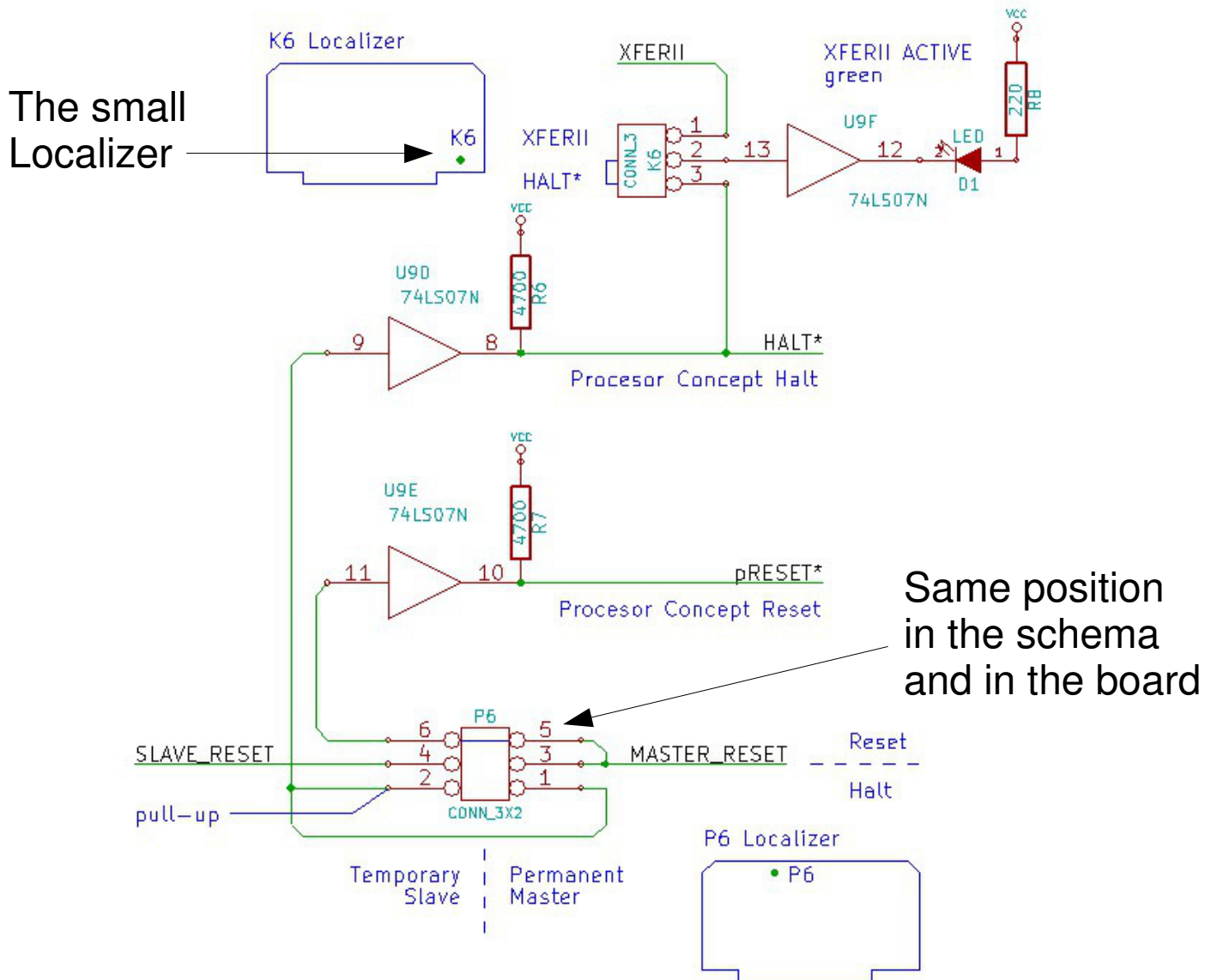
- Maximum processor frequency (if 68000-P8 no more than 16MHz)
- Maximum PHI signal in the S-100 bus (if other boards use this signal)

The Localizer (The large and complete one)

U29



Reset and Halt Configuration (P6)



P6- Permanent or Temporary Master Configuration

Significance of the straps:

4-6, Temporary Master Processor Reset

5-6, Alone Master Processor Reset

4-2, Temporary Halt (not tested)

3-1, Permanent Master Halt (not tested)

Other combinations have not signification

4-3, FORBIDDEN (possible short circuit)

If pins 1 or 2 are not select, a pull-up resistor is needed.

Table of configuration points

The Table of configuration points it is resumed in 5 fields:

Reference (ref)

It is an alphabetical list of all of the configurable point with strap that is possible to find in this board

The configurable points

It is a short explanation of the objective of this point. In bold character are remarked the normal points that might be used to fast commissioning for a Temporary or a Permanent Master. **But** what the board at the end can do it is only responsibility of the commissionist.

The Master Alone

It is the list of normal points to choose for an Alone Master configuration. This was the first line of the analysis objective of the workbench. Software will be the same of a Temporary Master but it never return the bus to any permanent master.

See page

It is a list of page in this document that will find a more extended explanation of the functionality of this point.

Care

This column is a way to find the experience, as a feeling, that how it seems difficult and what importance have it to the first start:

Vital – must be select a correct election to the first start.

Some care – will be selected with some care to the first start.

Do not care – is not important question to the first start.

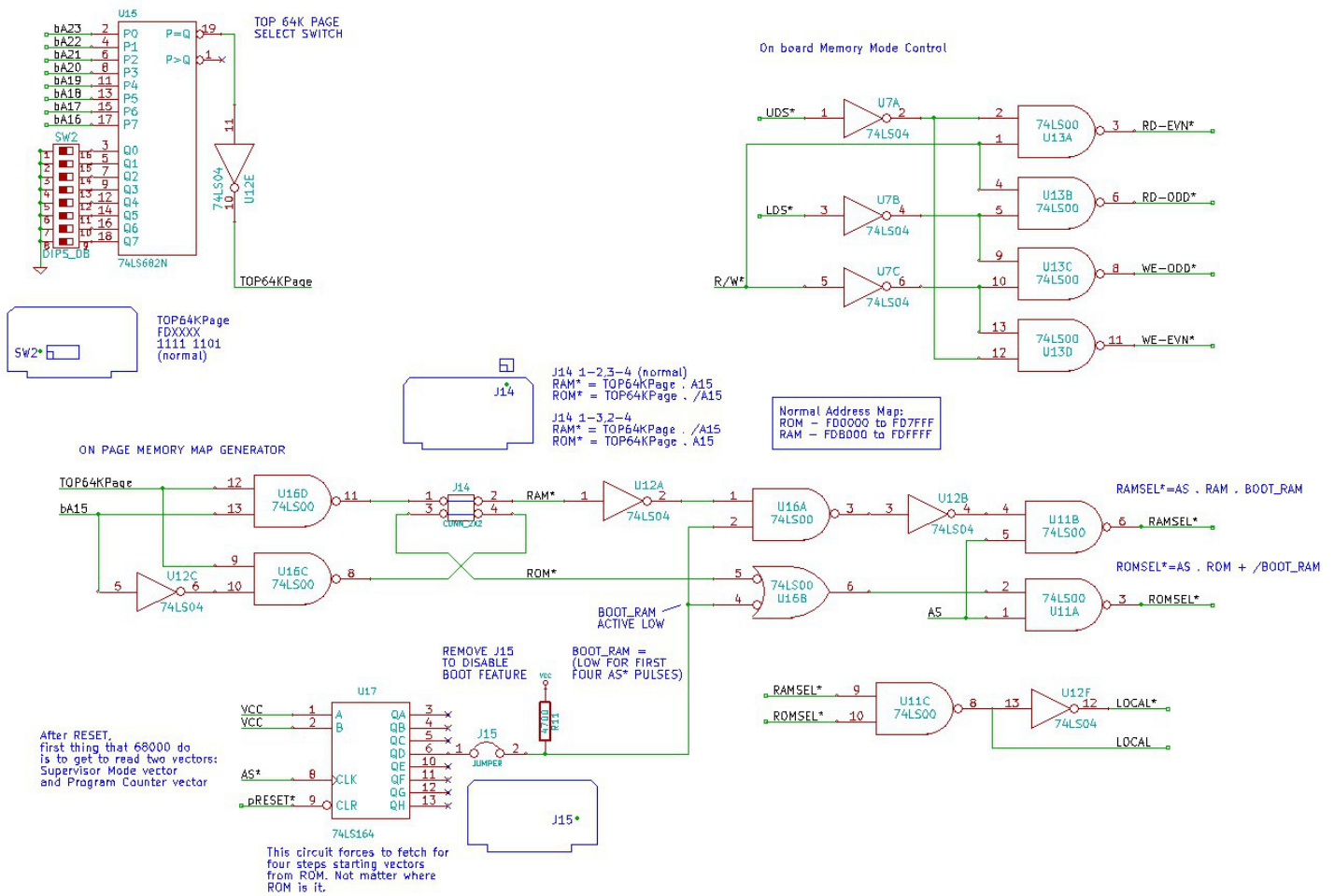
ref	Configurable points 68000 CPU V1.03 board	Master Alone	See page	Care
J9	Short , if WATCHDOG TIMER must be enabled	Short	TIMER	do not care
J10	Short , enables Auto-Vector VAP* in 68000 procesor	Short	XFERII	some difere nces
J13	Vectored Interrupt Inputs, Not Straps needed	VI1* 11-12	CPU	do not care
J14	1-2 & 3-4 Veritcal, RAM FD8000 > ROM FD0000	1-2 & 3-4	ADDR	Vital
J15	Short , Starting Vectors at Zero Page	Short	ADDR	Vital
JB1	1-2 & 3-4 : VCC and A15 in EEPROM conf.	1-2 & 3-4	MEMORY	Vital
JB2	3-4 & 7-8 : A14 and A15 in RAM configuration	3-4 & 7-8	MEMORY	Vital
JP1	1- GND; 2- 0V_A pin 20			do not care
JP2	1- GND; 2- 0V_B pin 53			do not care
JP3	1- GND; 2- 0V_C pin 70			do not care
JP4	Open : SBSD* not used if not permanent master	Open	GRANT	Vital
JP5	Open : NDEF1 not used	Open	GRANT	Vital
JP6	Short , if RESET* will be generated on board	Open	RESET	do not care

JP8	Short , if SLAVE_CLR* will be generated on board	Short	RESET	do not care
JP9	Short , if POC* will be generated on board	Short	RESET	do not care
JP10	Reset for Alone Master without SMB in system, but not pull-up resistor exist in the board. Must be placed a 1KΩ resistor, from C106 to VCC, if needed. Do not need if a SMB board is in the system	Open	RESET	do not care
JP11	Short , if enables equation [MWRT= sOUT* & pWR] see IEEE696-83 Table 5. Be sure this equation is not enabled in other place of the system like the System Monitor Board (SMB)	Short	TMA	Some care, not Vital
JP12	Short , enables CLOCK 2MHz to the S100 bus. Be sure this signal is not enabled in other place of the system	Short	XFERI	Some care, not Vital
JP13	Normal short . No other utilities for pin 3 of U34 are designed in the document	Short	TIMER	do not care
JP14	Open , Short only in a permanent master configuration. Enabling that a HOLD* request will activate a Bus Request* in 68000 processor	Open	GRANT	Vital
JP15	Open , Short only in a permanent master configuration. pHLDA output is send to relinquish the bus.	Open	GRANT	Vital

JP16	Open , Short only in a permanent master configuration. pHLDA output is send to relinquish the bus by the NDEF3 pin 66 line.	Open	GRA NT	Vital
K1	2-3 : Enables XFERI buffer permanently. 3-4 in a temporary master when grants the bus from the permanent master.	2-3 only	TMA	Vital
K2	1-3 special bridge : Enables XFERII buffer permanently. 2-3 in a temporary master when grants the bus from the permanent master.	1-3 special bridge	TMA	Vital
K3	2-3 used as MEE error interrupt. 1-2 is used as NMI interrupt.	1-2 as NMI	CPU	Some care, not Vital
K4	2-3 if XFERII enabled permanently. 1-2: when permanent master.	2-3 as XFERI I	TMA	Vital
K5	1-2 LOCAL* internal control, 2-3 DODSB* by S100 bus control	1-2 LOCAL	DATA BUF FER	Vital
K6	2-3 : LED D1 shows HALT* state active when it shines. 1-2: Short, as D1 as XFERII monitor	2-3 HALT*	RES ET	do not care
K7	2-3 : LED D2 shows bA20 line state active when it not shines. 1-2: Short, as D2 as FC2 monitor	2-3 bA20	WILC OX	do not care

K8	1-2: HLDA* or 2-3: NDEF3, as temporary master	do not care	TMA	do not care
K9	1-2: NDEF1 or 2-3: HOLD*, as temporary master	do not care	TMA	do not care
P2	WAIT-STATES: J4 7-8 and J6 11-12 are tested as must stable combination.	J4, J6	TIME R	Some care, not Vital
P5	1-2: TMAXPU for pull-up resistor. 5-6 for TMA1* activation from de permanent master. Adequate monitor program is needed in permanent master.	2 to GND	TMA	Vital
P6	4-6: As temporary master Reset from TMA circuit. 1-3: As HALT from bus RESET* line.	5-6	RESET	Vital
SW2	TOP 64K PAGE FDXXXX (1111 1101)	1111 1101	ADDRESS	Vital

ADDR PAGE



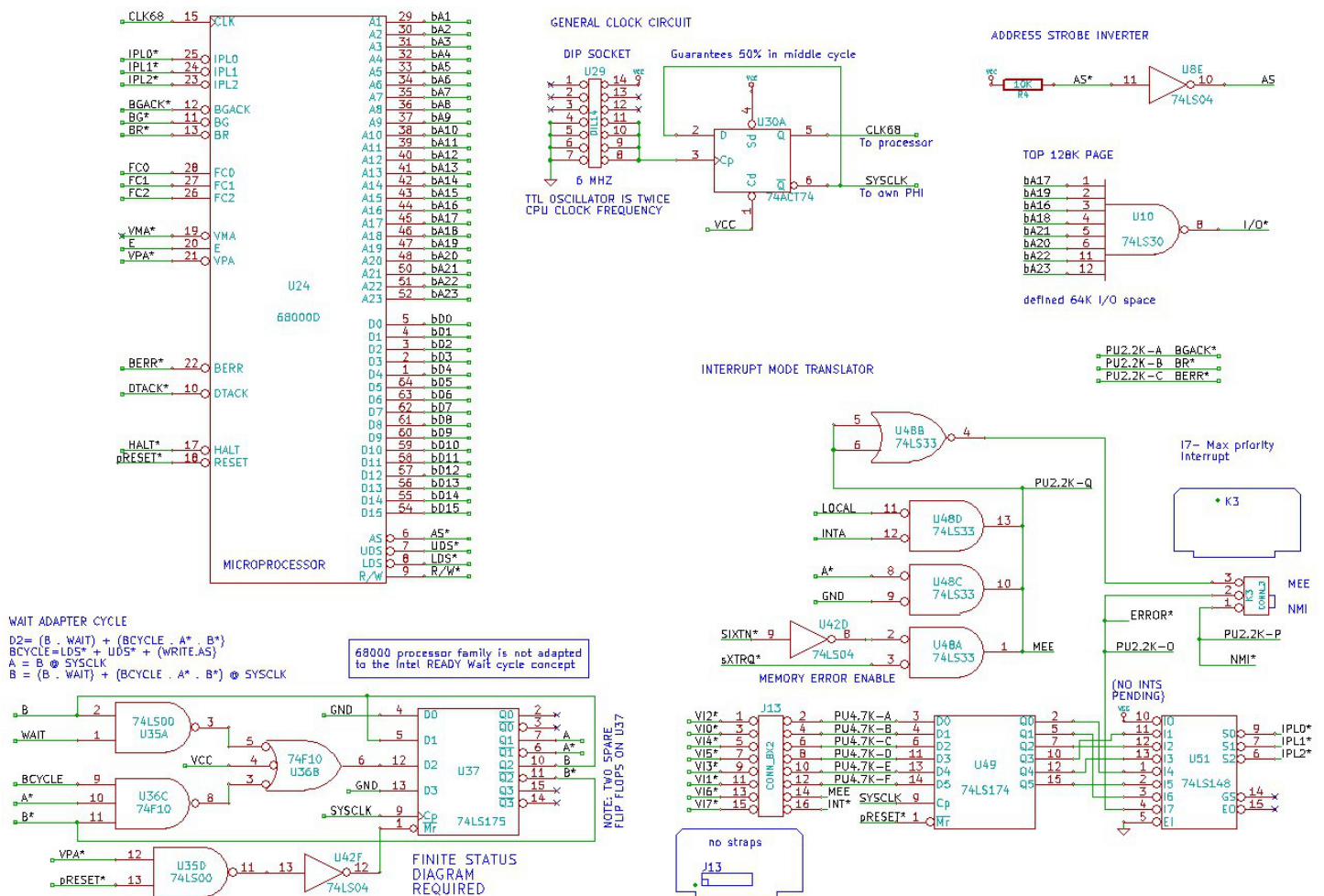
This circuitry is elaborated to determine the memory location. This must be in strict relationship to the monitor software addresses expressed in the program monitor.

SW2 – Selects the initial point where the memory map is extended, from this to up. This is FD0000 in hexadecimal format.

J14 – This strap offers the possibility to define which kind of memory is starting in the defined map (ROM or RAM). ROM is defined in FD0000 and RAM is defined in FD8000.

J15 – Disables the determined location of the memory map during the first four cycles after reset. 68000 processor wake-up in a different manner than Intel processors. It needs to find his initial vectors at address space started in 000000.

CPU PAGE



This circuitry determines how the 68000 processor cycles, interrupts and IO space reacts following the IEEE696 protocols in the S-100 bus.

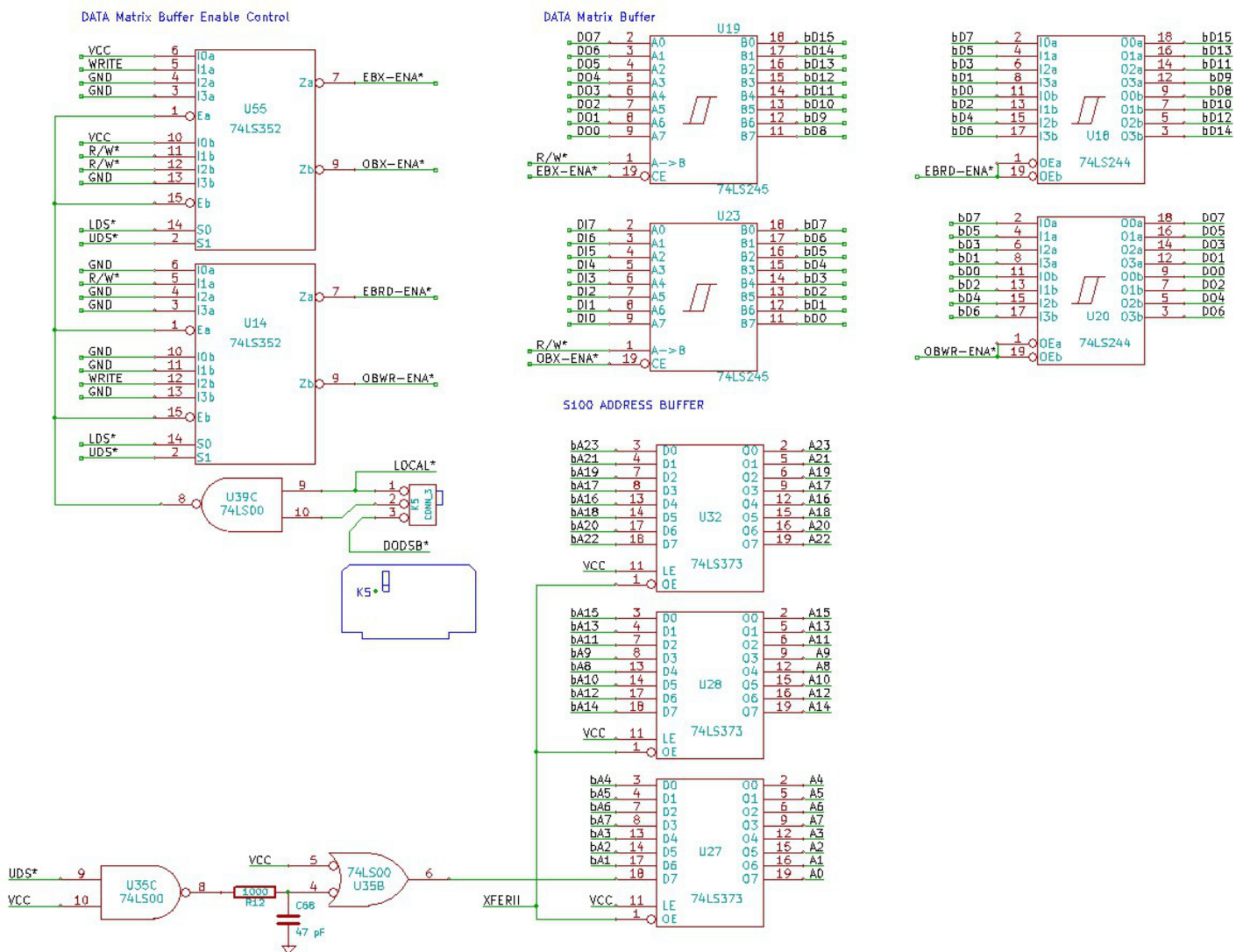
J13 – Working as a master, the processor must have the ability to attend the requested interruptions from the slaves. Those vectored interrupts (VI*) are conducted and translated by priority to the tree IPL0-2 Interrupt Priority Level input processor lines.

K3 – The must priority level input point to the processor is reserved to a special kind of interrupts in two possibilities:

- NMI* is used as input from the S-100 to request to the CPU that a fatality is occurred in the system.
- MEE is used as indication of internal fatalities.

In other hand, those tree pins can be united together because pull-up resistors additions results 1K1 ohms.

DATA BUFFER PAGE

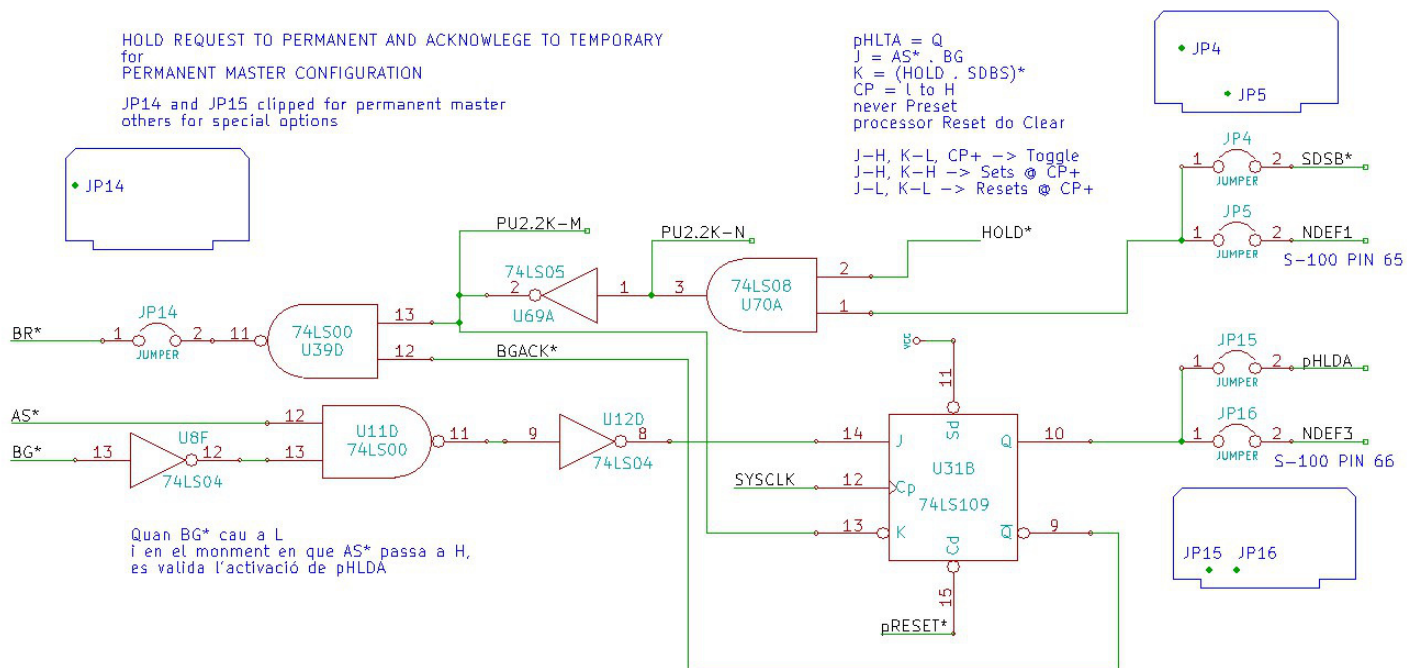


The “Data Matrix Buffer Enable Control” is used in this board as a way to adapt the 16 bits data bus to the Temporary Master Access protocol of the S-100 norm.

K5 – DODSB* is the inhibition signal from a Temporary Master to inhibit outputs of the Permanent Master. When this board is used as permanent master this signal becomes low when the bus is relinquished. This low is input to the pin 10 of the U39C and in consequence the output pin 8 becomes high. This high disables U55 and U14 and for that the entire data buffer becomes in silence (tree-state)

When this functionality is not used, pins 9 and 10 of U39C must be shared together in the normal control by LOCAL* signal.

GRANT PAGE



If the CPU 68000 board is not working as Permanent Master no straps are needed in no place of this schematic. If no master monitor is programmed for this circuit it does not have any utility.

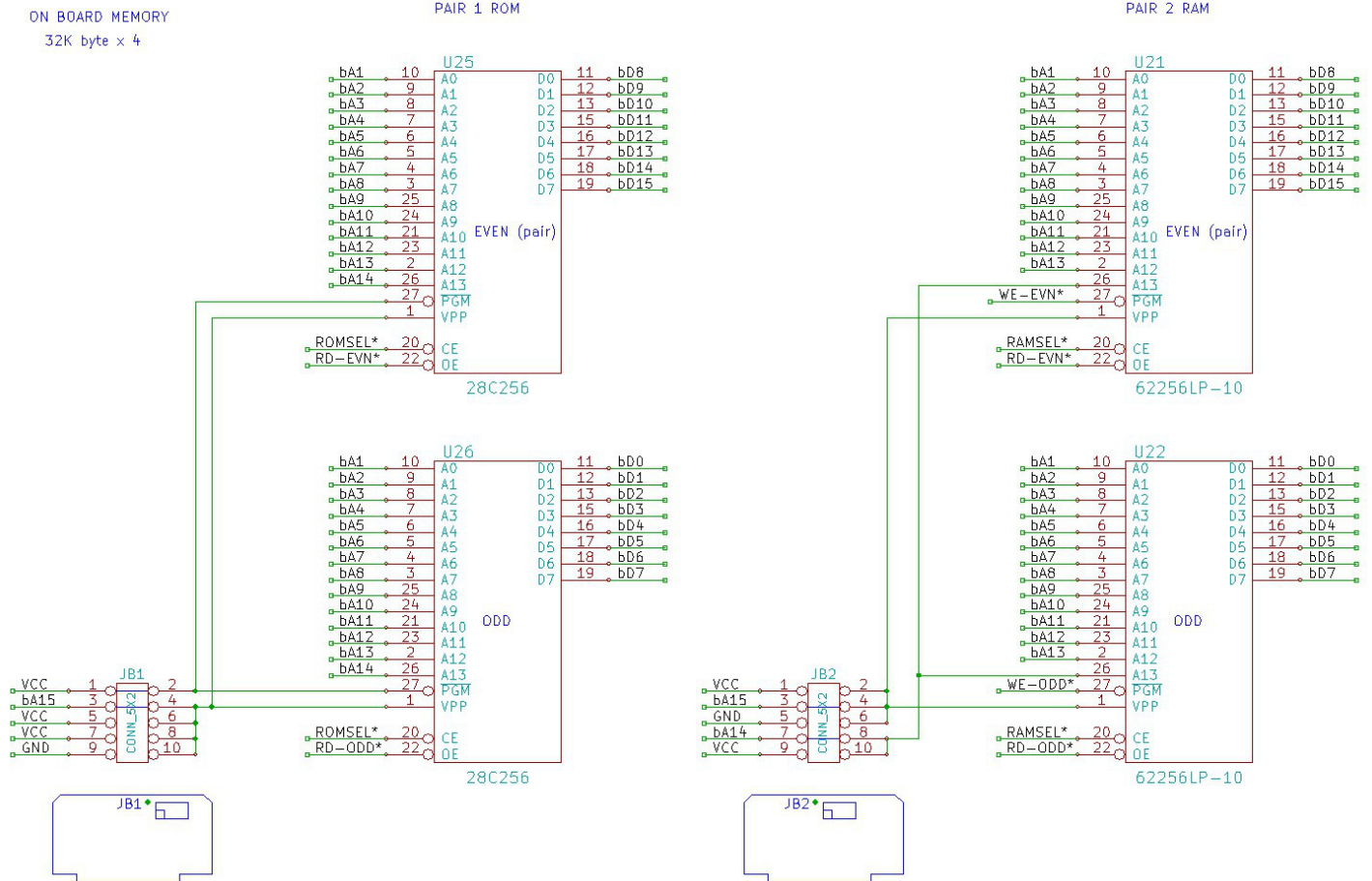
J14 – Enables a temporary master request arrives to the BR* bus request processor input.

JP4 – Enables the inhibition of status signal from the permanent master.

JP15 – Enables to a temporary master to be able to grant the bus.

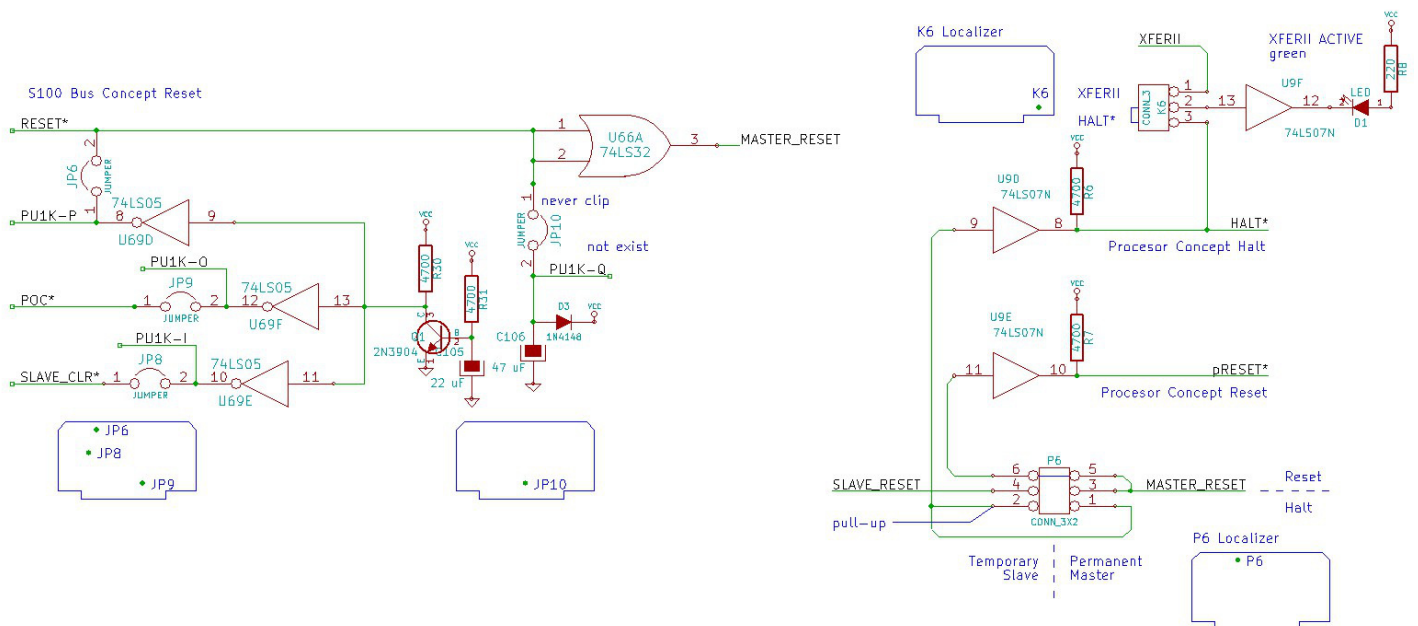
JP5 and **JP16** – Does similar functions of JP4 and JP15 but by alternative not defined pins of the bus.

MEMORY PAGE



The combination of pin straps of JB1 and JB2 are resolved only for the related memory integrated circuits. Other combinations must be optimized by the strict analysis from the data sheet of the alternative memory chips.

RESET PAGE



This is a modified version of the original reset circuitry of this board. This circuit pretends to wake-up the processor in a couple of situations: When a significant variation o activation of the power supply occurs or when the CPU grant the bus.

JP6 – Q1 transistor waits some time to be active until C105 rises to a voltage of 0.6V nominally in the Base-Emitter union. This will be only useful when the power supply powers up for a first time. This operation can be omitted if a System Monitor Board is plugged to the bus. In consequence the same consideration is applied to **JP8** and **JP9**.

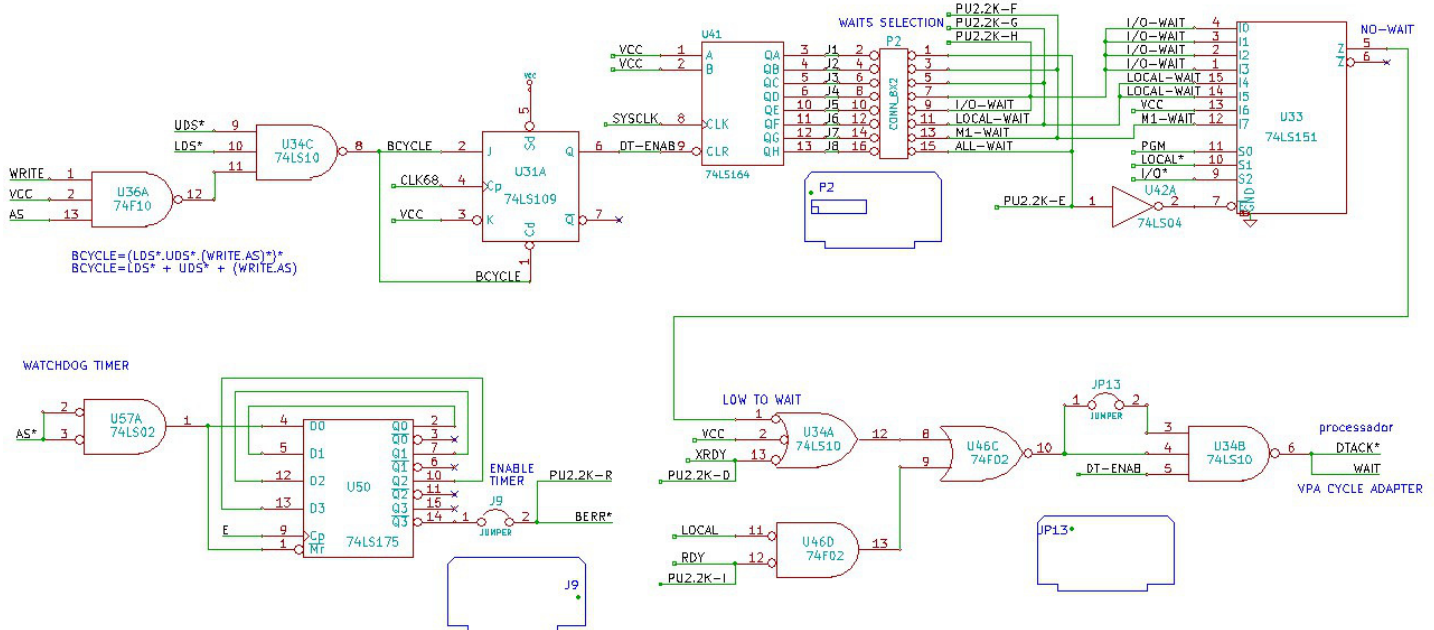
JP10 – This passive reset circuit generator assures a fast discharge through D3 diode. Unfortunately PU1K-Q, same of other PU1K references are not placed in the board. The only way to use this circuit is by a SMB plugged in the bus and it is enough able to do this passive reset functionality.

P6 – This is yet described in the first section of this document. This strap have two input (slave_master and master_reset) and two output (HALT* and pRESET*)

TIMER PAGE

WAIT-STATES GENERATOR

ADDED THE "-WAIT" SUFFIX TO THE U41A/P2/U33 CONNECTIONS TO DISTINGUISH THEM FROM OTHER BOARD NETNAMES - 3 JAN 2010 - AJL



This page describes the Wait-States Generator and the Watch-Doc Timer.

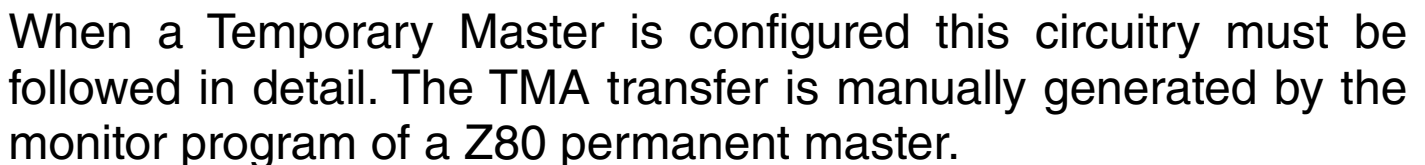
U41 counts a number of SYSCLK pulses until the S-100 bus cycle will be restarted.

P2 – J1 to J8 are an amount of wait-state possibilities that implies I/O-WAIT, LOCAL-WAIT and M1-WAIT. J1 and J8 are used to enable all kind of cycles at the same time. J2 and J7 enables fetch instruction M1-WAIT be delay. J3 and J6 enables a couple of delays of local-wait. J4 and J5 enables the I/O-WAIT delay configuration. IO and LOCAL are delayed by normal an important amount of clock pulses in this board.

JP13 – This is an option for an advanced and not defined functionality. This strap must be always be shorted.

J9 – In a large number of AS* cycles, if E is not rising up, the BERR* bus error processor control input will be activated, and the processor enter in a vectored exception procedure. What happens it depends of the software and the type of processor.

TMAC GENERATOR



P5 – Enables the start of the transfer. A low in pin 1 of U8A starts the transfer. Only one of the temporary masters can be placed 1-2 strap in the total system. An other strap indicates which master is this board. In this s100computer protocol only 4 temporary master are possibles.

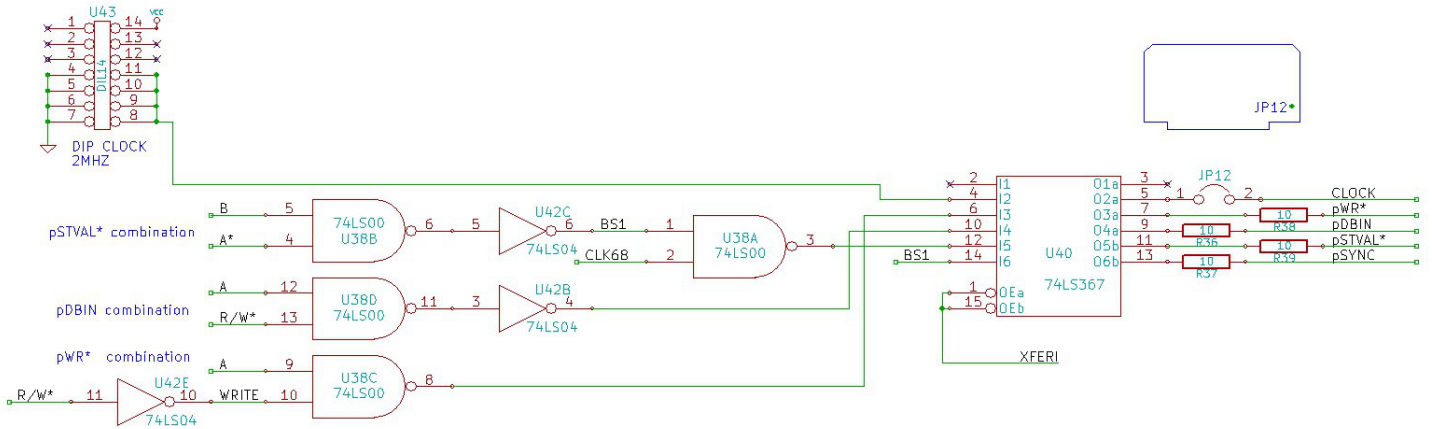
K1 and **K2** – Enables the XFERI and the XFERII status signals buffers in function of the “XFER” sequencer.

JP11 – Enables U46B function to go out to the bus.

In an Alone Master configuration this sequencer must be evolved to activate low the pin 1 of U72. This shines D4 and D5 led

XFERI PAGE

Bus State adapter, BS1 DRIVER & BUFFER controllet by XFERI

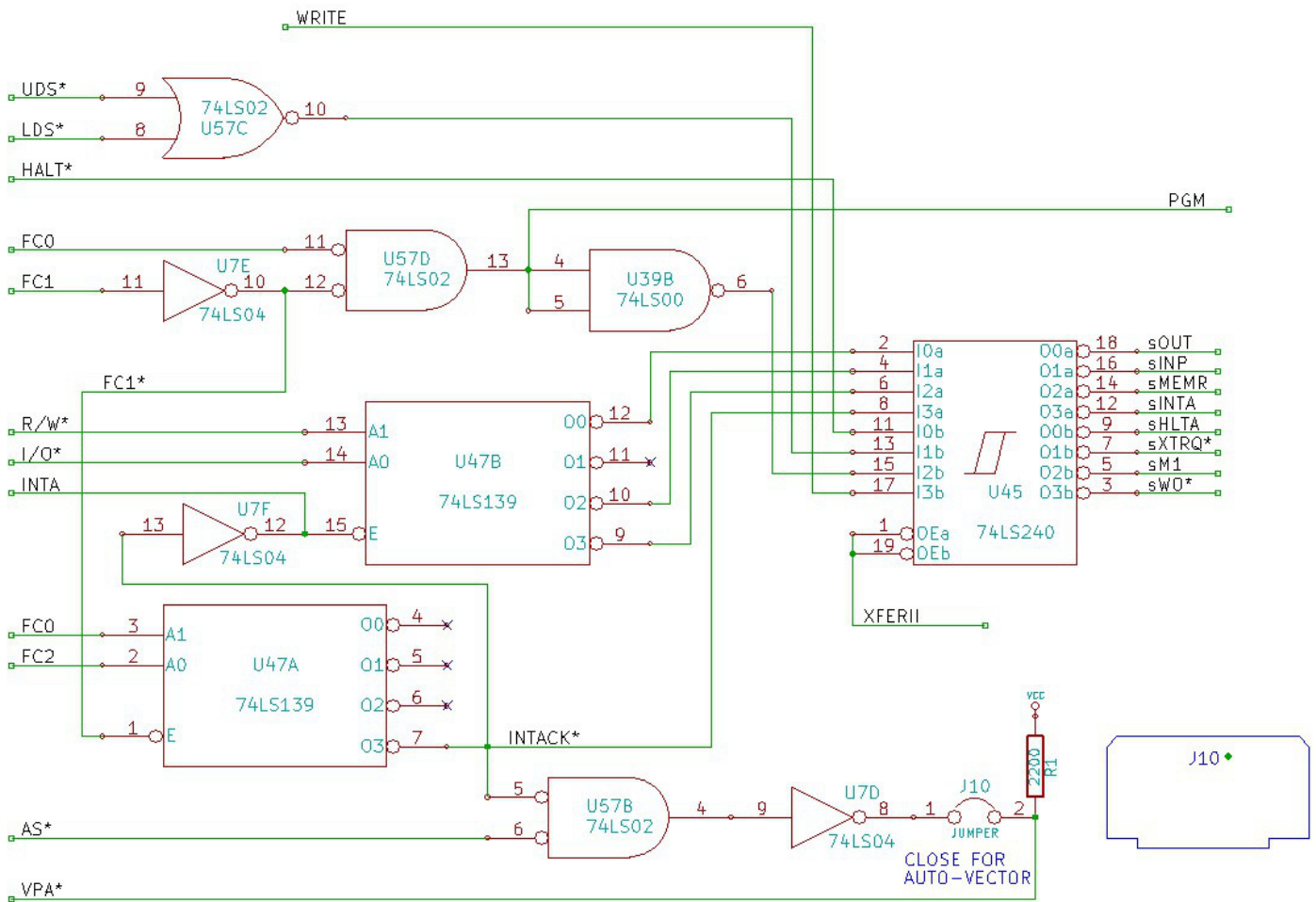


Not special things to comment is this schematic. Those are the final gates from the bus state cycle generator output and feedback signals (B,A*,A,R/W*)

P12 – Enables the 2MHz normalized frequency to be output to the bus. If the SMB board do that, this strap must not be placed.

XFERII PAGE

XFERII DRIVER & BUFFER

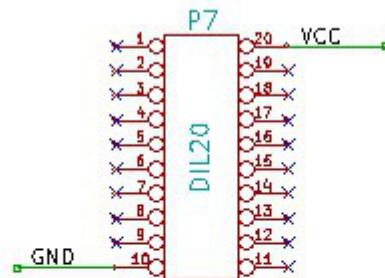


This circuitry translates the processor status signals to the IEEE696 norm. U45 becomes enable ever the temporary master grants the bus and will be inhibited when as permanent master is relinquishing the bus.

J10 – This Enables a VPA* to be active low when and interrupt acknowledge is generated by the processor. This is needed in the context of the monitor program and is not compiled how it is working in this document. It depends of the software.

WILCOX and PSU PAGE

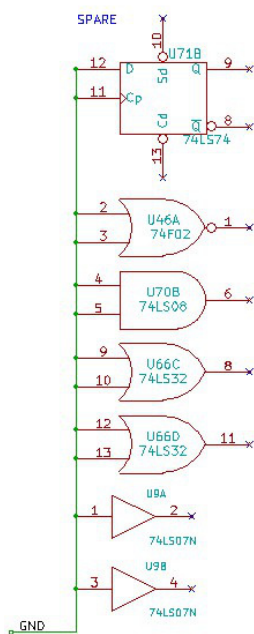
WILCOX TEST ZOCKET



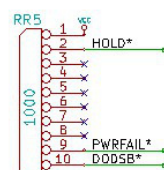
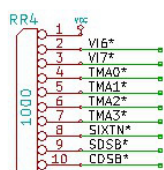
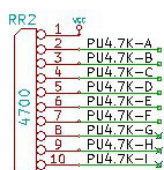
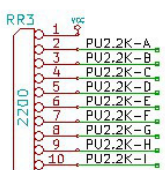
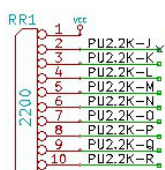
WILCOX TEST



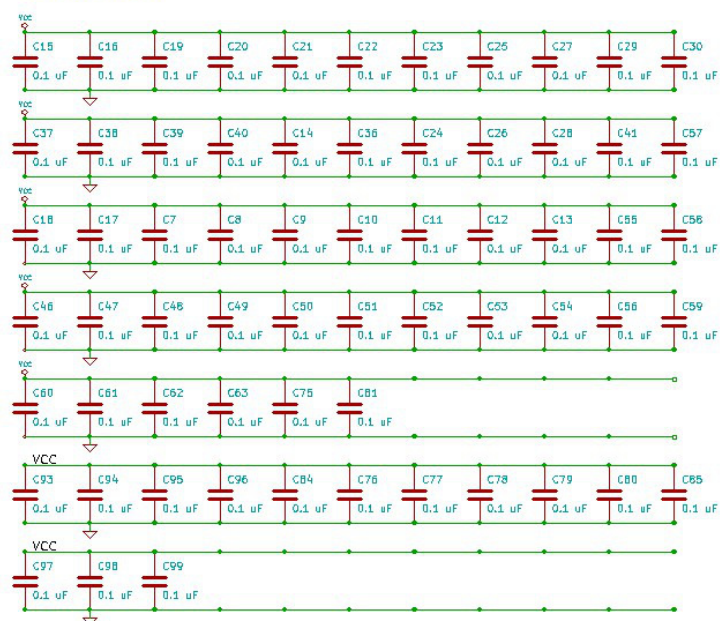
Requires the Wilcox book



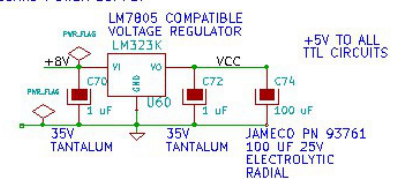
General Pull-Ups



DECOUPLING BATTERY



ON BOARD POWER SUPPLY



NOTE: FOR IMPROVED VOLTAGE REGULATION
C69 AND C70 CAN BE MUCH LARGER CAPACITANCE 47 uF AT 35V
C71 AND C72 CAN BE MUCH LARGER CAPACITANCE 10 uF AT 12V

CONCLUSION

Not being exhaustive in the description of the total functionalities of this board, this commissioning chart may do easy the first start of the microprocessor 68000 with the S100Computer program monitor.

By the way, seems to be difficult to construct a full system with several processor working together, one as a permanent master and others as temporary masters in the objective to rising an effective multiprocessor system.

But this compilation is realized to accomplishes a compromise with a user of the S100Computer community.

This beautiful board and his art of only TTL technologies gives to this writer a special glad to be commissioned this board until this stage. This quite two years that this board was stain in the workbench will be also used to discover some possibilities of the Standard-Hundred. Some discoveries like the Alone Master or other capabilities that is possible to be rising as Temporary Master gives an enormous satisfaction.

S100Computers community gives the opportunity to review this elemental technology and the inspiration to other advanced capabilities of the IEEE696-83 norm.

Anonymously

